

4. Verify that $y = \sin 3t + 2 \cos 3t$ is a solution to the initial value problem

$$2y'' + 18y = 0; \quad y(0) = 2, \quad y'(0) = 3.$$

Find the maximum of $|y(t)|$ for $-\infty < t < \infty$.

5. Verify that the exponentially damped sinusoid $y(t) = e^{-3t} \sin(\sqrt{3}t)$ is a solution to equation (3) if $F_{\text{ext}}(t) = 0$, $m = 1$, $b = 6$, and $k = 12$. What is the limit of this solution as $t \rightarrow \infty$?
6. An external force $F(t) = 2 \cos 2t$ is applied to a mass-spring system with $m = 1$, $b = 0$, and $k = 4$, which is initially at rest; i.e., $y(0) = 0$, $y'(0) = 0$. Verify that $y(t) = \frac{1}{2}t \sin 2t$ gives the motion of this spring. What will eventually (as t increases) happen to the spring?

In Problems 7–9, find a synchronous solution of the form $A \cos \Omega t + B \sin \Omega t$ to the given forced oscillator equation using the method of Example 4 to solve for A and B .

7. $y'' + 2y' + 4y = 5 \sin 3t$, $\Omega = 3$
 8. $y'' + 2y' + 5y = -50 \sin 5t$, $\Omega = 5$
 9. $y'' + 2y' + 4y = 6 \cos 2t + 8 \sin 2t$, $\Omega = 2$
10. Undamped oscillators that are driven at resonance have unusual (and nonphysical) solutions.

- (a) To investigate this, find the synchronous solution $A \cos \Omega t + B \sin \Omega t$ to the generic forced oscillator equation

$$(7) \quad my'' + by' + ky = \cos \Omega t.$$



- (b) Sketch graphs of the coefficients A and B , as functions of Ω , for $m = 1$, $b = 0.1$, and $k = 25$.



- (c) Now set $b = 0$ in your formulas for A and B and resketch the graphs in part (b), with $m = 1$, and $k = 25$. What happens at $\Omega = 5$? Notice that the amplitudes of the synchronous solutions grow without bound as Ω approaches 5.

- (d) Show directly, by substituting the form $A \cos \Omega t + B \sin \Omega t$ into equation (7), that when $b = 0$ there are no synchronous solutions if $\Omega = \sqrt{k/m}$.

- (e) Verify that $(2m\Omega)^{-1}t \sin \Omega t$ solves equation (7) when $b = 0$ and $\Omega = \sqrt{k/m}$. Notice that this nonsynchronous solution grows in time, without bound.

Clearly one cannot neglect damping in analyzing an oscillator forced at resonance, because otherwise the solutions, as shown in part (e), are nonphysical. This behavior will be studied later in this chapter.

4.2 Homogeneous Linear Equations: The General Solution

We begin our study of the linear second-order constant-coefficient differential equation

$$(1) \quad ay'' + by' + cy = f(t) \quad (a \neq 0)$$

with the special case where the function $f(t)$ is zero:

$$(2) \quad ay'' + by' + cy = 0.$$

This case arises when we consider mass-spring oscillators vibrating freely—that is, without external forces applied. Equation (2) is called the *homogeneous form* of equation (1); $f(t)$ is the “nonhomogeneity” in (1). (This nomenclature is not related to the way we used the term for first-order equations in Section 2.6.)

A look at equation (2) tells us that a solution of (2) must have the property that its second derivative is expressible as a linear combination of its first and zeroth derivatives.[†] This suggests that we try to find a solution of the form $y = e^{rt}$, since derivatives of e^{rt} are just constants times e^{rt} . If we substitute $y = e^{rt}$ into (2), we obtain

$$\begin{aligned} ar^2 e^{rt} + bre^{rt} + ce^{rt} &= 0, \\ e^{rt}(ar^2 + br + c) &= 0. \end{aligned}$$

[†]The zeroth derivative of a function is the function itself.

Because e^{rt} is never zero, we can divide by it to obtain

$$(3) \quad ar^2 + br + c = 0.$$

Consequently, $y = e^{rt}$ is a solution to (2) if and only if r satisfies equation (3). Equation (3) is called the **auxiliary equation** (also known as the **characteristic equation**) associated with the homogeneous equation (2).

Now the auxiliary equation is just a quadratic, and its roots are

$$r_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \quad \text{and} \quad r_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}.$$

When the discriminant, $b^2 - 4ac$, is positive, the roots r_1 and r_2 are real and distinct. If $b^2 - 4ac = 0$, the roots are real and equal. And when $b^2 - 4ac < 0$, the roots are complex conjugate numbers. We consider the first two cases in this section; the complex case is deferred to Section 4.3.

Example 1 Find a pair of solutions to

$$(4) \quad y'' + 5y' - 6y = 0.$$

Solution The auxiliary equation associated with (4) is

$$r^2 + 5r - 6 = (r - 1)(r + 6) = 0,$$

which has the roots $r_1 = 1$, $r_2 = -6$. Thus, e^t and e^{-6t} are solutions. ♦

Notice that the identically zero function, $y(t) \equiv 0$, is always a solution to (2). Furthermore, when we have a pair of solutions $y_1(t)$ and $y_2(t)$ to this equation, as in Example 1, we can construct an infinite number of other solutions by forming linear combinations:

$$(5) \quad y(t) = c_1 y_1(t) + c_2 y_2(t)$$

for any choice of the constants c_1 and c_2 . The fact that (5) is a solution to (2) can be seen by direct substitution and rearrangement:

$$\begin{aligned} ay'' + by' + cy &= a(c_1 y_1 + c_2 y_2)'' + b(c_1 y_1 + c_2 y_2)' + c(c_1 y_1 + c_2 y_2) \\ &= a(c_1 y_1'' + c_2 y_2'') + b(c_1 y_1' + c_2 y_2') + c(c_1 y_1 + c_2 y_2) \\ &= c_1 (ay_1'' + by_1' + cy_1) + c_2 (ay_2'' + by_2' + cy_2) \\ &= 0 + 0. \end{aligned}$$

The two “degrees of freedom” c_1 and c_2 in the combination (5) suggest that solutions to the differential equation (2) can be found meeting additional conditions, such as the initial conditions for the first-order equations in Chapter 1. But the presence of c_1 and c_2 leads one to anticipate that *two* such conditions, rather than just one, can be imposed. This is consistent with the mass–spring interpretation of equation (2), since predicting the motion of a mechanical system requires knowledge not only of the forces but also of the initial position $y(0)$ and velocity $y'(0)$ of the mass. A typical *initial value problem* for these second-order equations is given in the following example.

Example 2 Solve the initial value problem

$$(6) \quad y'' + 2y' - y = 0; \quad y(0) = 0, \quad y'(0) = -1.$$

Solution We will first find a pair of solutions as in the previous example. Then we will adjust the constants c_1 and c_2 in (5) to obtain a solution that matches the initial conditions on $y(0)$ and $y'(0)$. The auxiliary equation is

$$r^2 + 2r - 1 = 0.$$

Using the quadratic formula, we find that the roots of this equation are

$$r_1 = -1 + \sqrt{2} \quad \text{and} \quad r_2 = -1 - \sqrt{2}.$$

Consequently, the given differential equation has solutions of the form

$$(7) \quad y(t) = c_1 e^{(-1+\sqrt{2})t} + c_2 e^{(-1-\sqrt{2})t}.$$

To find the specific solution that satisfies the initial conditions given in (6), we first differentiate y as given in (7), then plug y and y' into the initial conditions of (6). This gives

$$y(0) = c_1 e^0 + c_2 e^0,$$

$$y'(0) = (-1 + \sqrt{2})c_1 e^0 + (-1 - \sqrt{2})c_2 e^0,$$

or

$$0 = c_1 + c_2,$$

$$-1 = (-1 + \sqrt{2})c_1 + (-1 - \sqrt{2})c_2.$$

Solving this system yields $c_1 = -\sqrt{2}/4$ and $c_2 = \sqrt{2}/4$. Thus,

$$y(t) = -\frac{\sqrt{2}}{4} e^{(-1+\sqrt{2})t} + \frac{\sqrt{2}}{4} e^{(-1-\sqrt{2})t}$$

is the desired solution. ♦

To gain more insight into the significance of the two-parameter solution form (5), we need to look at some of the properties of the second-order equation (2). First of all, there is an existence-and-uniqueness theorem for solutions to (2); it is somewhat like the corresponding Theorem 1 in Section 1.2 for first-order equations but updated to reflect the fact that *two* initial conditions are appropriate for *second*-order equations. As motivation for the theorem, suppose the differential equation (2) were *really easy*, with $b = 0$ and $c = 0$. Then $y'' = 0$ would merely say that the graph of $y(t)$ is simply a *straight line*, so it is uniquely determined by specifying a point on the line,

$$(8) \quad y(t_0) = Y_0,$$

and the slope of the line,

$$(9) \quad y'(t_0) = Y_1.$$

Theorem 1 states that conditions (8) and (9) suffice to determine the solution uniquely for the more general equation (2).

Existence and Uniqueness: Homogeneous Case

Theorem 1. For any real numbers a ($\neq 0$), b , c , t_0 , Y_0 , and Y_1 , there exists a unique solution to the initial value problem

$$(10) \quad ay'' + by' + cy = 0; \quad y(t_0) = Y_0, \quad y'(t_0) = Y_1.$$

The solution is valid for all t in $(-\infty, +\infty)$.

Note in particular that if a solution $y(t)$ and its derivative vanish simultaneously at a point t_0 (i.e., $Y_0 = Y_1 = 0$), then $y(t)$ must be the identically zero solution.

In this section and the next, we will construct explicit solutions to (10), so the question of *existence* of a solution is not really an issue. It is extremely valuable to know, however, that the solution is *unique*. The proof of uniqueness is rather different from anything else in this chapter, so we defer it to Chapter 13.[†]

Now we want to use this theorem to show that, given two solutions $y_1(t)$ and $y_2(t)$ to equation (2), we can always find values of c_1 and c_2 so that $c_1y_1(t) + c_2y_2(t)$ meets specified initial conditions in (10) and therefore is the (unique) solution to the initial value problem. But we need to be a little more precise; if, for example, $y_2(t)$ is simply the identically zero solution, then $c_1y_1(t) + c_2y_2(t) = c_1y_1(t)$ actually has only *one* constant and cannot be expected to satisfy *two* conditions. Furthermore, if $y_2(t)$ is simply a constant multiple of $y_1(t)$ —say, $y_2(t) = \kappa y_1(t)$ —then again $c_1y_1(t) + c_2y_2(t) = (c_1 + \kappa c_2)y_1(t) = Cy_1(t)$ actually has only one constant. The condition we need is *linear independence*.

Linear Independence of Two Functions

Definition 1. A pair of functions $y_1(t)$ and $y_2(t)$ is said to be **linearly independent on the interval I** if and only if neither of them is a constant multiple of the other on all of I .^{††} We say that y_1 and y_2 are **linearly dependent on I** if one of them *is* a constant multiple of the other on all of I .

Representation of Solutions to Initial Value Problem

Theorem 2. If $y_1(t)$ and $y_2(t)$ are any two solutions to the differential equation (2) that are linearly independent on $(-\infty, \infty)$, then unique constants c_1 and c_2 can always be found so that $c_1y_1(t) + c_2y_2(t)$ satisfies the initial value problem (10) on $(-\infty, \infty)$.

The proof of Theorem 2 will be easy once we establish the following technical lemma.

A Condition for Linear Dependence of Solutions

Lemma 1. For any real numbers a ($\neq 0$), b , and c , if $y_1(t)$ and $y_2(t)$ are any two solutions to the differential equation (2) on $(-\infty, \infty)$ and if the equality

$$(11) \quad y_1(\tau)y_2'(\tau) - y_1'(\tau)y_2(\tau) = 0$$

holds at *any* point τ , then y_1 and y_2 are linearly dependent on $(-\infty, \infty)$. (The expression on the left-hand side of (11) is called the *Wronskian* of y_1 and y_2 at the point τ ; see Problem 34 on page 164.)

[†]All references to Chapters 11–13 refer to the expanded text, *Fundamentals of Differential Equations and Boundary Value Problems*, 7th ed.

^{††}This definition will be generalized to three or more functions in Problem 35 and Chapter 6.

Proof of Lemma 1. *Case 1.* If $y_1(\tau) \neq 0$, then let κ equal $y_2(\tau)/y_1(\tau)$ and consider the solution to (2) given by $y(t) = \kappa y_1(t)$. It satisfies the same “initial conditions” at $t = \tau$ as does $y_2(t)$:

$$y(\tau) = \frac{y_2(\tau)}{y_1(\tau)} y_1(\tau) = y_2(\tau); \quad y'(\tau) = \frac{y_2(\tau)}{y_1(\tau)} y_1'(\tau) = y_2'(\tau),$$

where the last equality follows from (11). By uniqueness, $y_2(t)$ must be the same function as $\kappa y_1(t)$ on I .

Case 2. If $y_1(\tau) = 0$ but $y_1'(\tau) \neq 0$, then (11) implies $y_2(\tau) = 0$. Let $\kappa = y_2'(\tau)/y_1'(\tau)$. Then the solution to (2) given by $y(t) = \kappa y_1(t)$ (again) satisfies the same “initial conditions” at $t = \tau$ as does $y_2(t)$:

$$y(\tau) = \frac{y_2'(\tau)}{y_1'(\tau)} y_1(\tau) = 0 = y_2(\tau); \quad y'(\tau) = \frac{y_2'(\tau)}{y_1'(\tau)} y_1'(\tau) = y_2'(\tau).$$

By uniqueness, then, $y_2(t) = \kappa y_1(t)$ on I .

Case 3. If $y_1(\tau) = y_1'(\tau) = 0$, then $y_1(t)$ is a solution to the differential equation (2) satisfying the initial conditions $y_1(\tau) = y_1'(\tau) = 0$; but $y(t) \equiv 0$ is the *unique* solution to this initial value problem. Thus, $y_1(t) \equiv 0$ [and is a constant multiple of $y_2(t)$]. ♦

Proof of Theorem 2. We already know that $y(t) = c_1 y_1(t) + c_2 y_2(t)$ is a solution to (2); we must show that c_1 and c_2 can be chosen so that

$$y(t_0) = c_1 y_1(t_0) + c_2 y_2(t_0) = Y_0$$

and

$$y'(t_0) = c_1 y_1'(t_0) + c_2 y_2'(t_0) = Y_1.$$

But simple algebra shows these equations have the solution[†]

$$c_1 = \frac{Y_0 y_2'(t_0) - Y_1 y_2(t_0)}{y_1(t_0) y_2'(t_0) - y_1'(t_0) y_2(t_0)} \quad \text{and} \quad c_2 = \frac{Y_1 y_1(t_0) - Y_0 y_1'(t_0)}{y_1(t_0) y_2'(t_0) - y_1'(t_0) y_2(t_0)}$$

as long as the denominator is nonzero, and the technical lemma assures us that this condition is met. ♦

Now we can honestly say that if y_1 and y_2 are linearly independent solutions to (2) on $(-\infty, +\infty)$, then (5) is a **general solution**, since *any* solution $y_g(t)$ of (2) can be expressed in this form; simply pick c_1 and c_2 so that $c_1 y_1 + c_2 y_2$ matches the value and the derivative of y_g at *any point*. By uniqueness, $c_1 y_1 + c_2 y_2$ and y_g have to be the same function. See Figure 4.6 on page 162.

How do we *find* a general solution for the differential equation (2)? We already know the answer if the roots of the auxiliary equation (3) are real and distinct because clearly $y_1(t) = e^{r_1 t}$ is not a constant multiple of $y_2(t) = e^{r_2 t}$ if $r_1 \neq r_2$.

Distinct Real Roots

If the auxiliary equation (3) has distinct real roots r_1 and r_2 , then both $y_1(t) = e^{r_1 t}$ and $y_2(t) = e^{r_2 t}$ are solutions to (2) and $y(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}$ is a general solution.

[†]To solve for c_1 , for example, multiply the first equation by $y_2'(t_0)$ and the second by $y_2(t_0)$ and subtract.

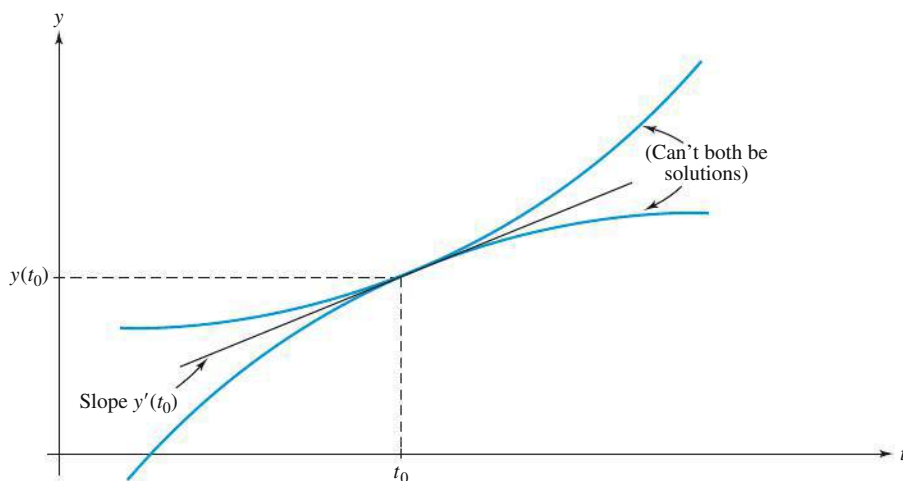


Figure 4.6 $y(t_0), y'(t_0)$ determine a *unique* solution.

When the roots of the auxiliary equation are equal, we only get one nontrivial solution, $y_1 = e^{rt}$. To satisfy *two* initial conditions, $y(t_0)$ and $y'(t_0)$, then we will need a second, linearly independent solution. The following rule is the key to finding a second solution.

Repeated Root

If the auxiliary equation (3) has a repeated root r , then both $y_1(t) = e^{rt}$ and $y_2(t) = te^{rt}$ are solutions to (2), and $y(t) = c_1e^{rt} + c_2te^{rt}$ is a general solution.

We illustrate this result before giving its proof.

Example 3 Find a solution to the initial value problem

$$(12) \quad y'' + 4y' + 4y = 0; \quad y(0) = 1, \quad y'(0) = 3.$$

Solution The auxiliary equation for (12) is

$$r^2 + 4r + 4 = (r + 2)^2 = 0.$$

Because $r = -2$ is a double root, the rule says that (12) has solutions $y_1 = e^{-2t}$ and $y_2 = te^{-2t}$. Let's confirm that $y_2(t)$ is a solution:

$$\begin{aligned} y_2(t) &= te^{-2t}, \\ y_2'(t) &= e^{-2t} - 2te^{-2t}, \\ y_2''(t) &= -2e^{-2t} - 2e^{-2t} + 4te^{-2t} = -4e^{-2t} + 4te^{-2t}, \\ y_2'' + 4y_2' + 4y_2 &= -4e^{-2t} + 4te^{-2t} + 4(e^{-2t} - 2te^{-2t}) + 4te^{-2t} = 0. \end{aligned}$$

Further observe that e^{-2t} and te^{-2t} are linearly independent since neither is a constant multiple of the other on $(-\infty, \infty)$. Finally, we insert the general solution $y(t) = c_1e^{-2t} + c_2te^{-2t}$ into the initial conditions,

$$\begin{aligned} y(0) &= c_1e^0 + c_2(0)e^0 = 1, \\ y'(0) &= -2c_1e^0 + c_2e^0 - 2c_2(0)e^0 = 3, \end{aligned}$$

and solve to find $c_1 = 1, c_2 = 5$. Thus $y = e^{-2t} + 5te^{-2t}$ is the desired solution. ♦

Why is it that $y_2(t) = te^{rt}$ is a solution to the differential equation (2) when r is a double root (and not otherwise)? In later chapters we will see a theoretical justification of this rule in very general circumstances; for present purposes, though, simply note what happens if we substitute y_2 into the differential equation (2):

$$\begin{aligned}y_2(t) &= te^{rt}, \\y_2'(t) &= e^{rt} + rte^{rt}, \\y_2''(t) &= re^{rt} + re^{rt} + r^2te^{rt} = 2re^{rt} + r^2te^{rt}, \\ay_2'' + by_2' + cy_2 &= [2ar + b]e^{rt} + [ar^2 + br + c]te^{rt}.\end{aligned}$$

Now if r is a root of the auxiliary equation (3), the expression in the second brackets is zero. However, *if r is a double root, the expression in the first brackets is zero also:*

$$(13) \quad r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-b \pm (0)}{2a};$$

hence, $2ar + b = 0$ for a double root. In such a case, then, y_2 is a solution.

The method we have described for solving homogeneous linear second-order equations with constant coefficients applies to any order (even first-order) homogeneous linear equations with constant coefficients. We give a detailed treatment of such higher-order equations in Chapter 6. For now, we will be content to illustrate the method by means of an example. We remark briefly that a homogeneous linear n th-order equation has a general solution of the form

$$y(t) = c_1y_1(t) + c_2y_2(t) + \cdots + c_ny_n(t),$$

where the individual solutions $y_i(t)$ are “linearly independent.” By this we mean that no y_i is expressible as a linear combination of the others; see Problem 35 on page 164.

Example 4 Find a general solution to

$$(14) \quad y''' + 3y'' - y' - 3y = 0.$$

Solution If we try to find solutions of the form $y = e^{rt}$, then, as with second-order equations, we are led to finding roots of the auxiliary equation

$$(15) \quad r^3 + 3r^2 - r - 3 = 0.$$

We observe that $r = 1$ is a root of the above equation, and dividing the polynomial on the left-hand side of (15) by $r - 1$ leads to the factorization

$$(r - 1)(r^2 + 4r + 3) = (r - 1)(r + 1)(r + 3) = 0.$$

Hence, the roots of the auxiliary equation are 1, -1 , and -3 , and so three solutions of (14) are e^t , e^{-t} , and e^{-3t} . The linear independence of these three exponential functions is proved in Problem 36. A general solution to (14) is then

$$(16) \quad y(t) = c_1e^t + c_2e^{-t} + c_3e^{-3t}. \quad \blacklozenge$$

So far we have seen only exponential solutions to the linear second-order constant coefficient equation. You may wonder where the vibratory solutions that govern mass–spring oscillators are. In the next section, it will be seen that they arise when the solutions to the auxiliary equation are complex.

4.2 EXERCISES

In Problems 1–12, find a general solution to the given differential equation.

1. $2y'' + 7y' - 4y = 0$
2. $y'' + 6y' + 9y = 0$
3. $y'' + 5y' + 6y = 0$
4. $y'' - y' - 2y = 0$
5. $y'' + 8y' + 16y = 0$
6. $y'' - 5y' + 6y = 0$
7. $6y'' + y' - 2y = 0$
8. $z'' + z' - z = 0$
9. $4y'' - 4y' + y = 0$
10. $y'' - y' - 11y = 0$
11. $4w'' + 20w' + 25w = 0$
12. $3y'' + 11y' - 7y = 0$

In Problems 13–20, solve the given initial value problem.

13. $y'' + 2y' - 8y = 0$; $y(0) = 3$, $y'(0) = -12$
14. $y'' + y' = 0$; $y(0) = 2$, $y'(0) = 1$
15. $y'' - 4y' + 3y = 0$; $y(0) = 1$, $y'(0) = 1/3$
16. $y'' - 4y' - 5y = 0$; $y(-1) = 3$, $y'(-1) = 9$
17. $y'' - 6y' + 9y = 0$; $y(0) = 2$, $y'(0) = 25/3$
18. $z'' - 2z' - 2z = 0$; $z(0) = 0$, $z'(0) = 3$
19. $y'' + 2y' + y = 0$; $y(0) = 1$, $y'(0) = -3$
20. $y'' - 4y' + 4y = 0$; $y(1) = 1$, $y'(1) = 1$

21. First-Order Constant-Coefficient Equations.

- (a) Substituting $y = e^{rt}$, find the auxiliary equation for the first-order linear equation

$$ay' + by = 0,$$

where a and b are constants with $a \neq 0$.

- (b) Use the result of part (a) to find the general solution.

In Problems 22–25, use the method described in Problem 21 to find a general solution to the given equation.

22. $3y' - 7y = 0$
23. $5y' + 4y = 0$
24. $3z' + 11z = 0$
25. $6w' - 13w = 0$

26. Boundary Value Problems. When the values of a solution to a differential equation are specified at two different points, these conditions are called **boundary conditions**. (In contrast, initial conditions specify the values of a function and its derivative at the same point.) The purpose of this exercise is to show that for boundary value problems there is no existence–uniqueness theorem that is analogous to Theorem 1. Given that every solution to

$$(17) \quad y'' + y = 0$$

is of the form

$$y(t) = c_1 \cos t + c_2 \sin t,$$

where c_1 and c_2 are arbitrary constants, show that

- (a) There is a unique solution to (17) that satisfies the boundary conditions $y(0) = 2$ and $y(\pi/2) = 0$.

- (b) There is no solution to (17) that satisfies $y(0) = 2$ and $y(\pi) = 0$.

- (c) There are infinitely many solutions to (17) that satisfy $y(0) = 2$ and $y(\pi) = -2$.

In Problems 27–32, use Definition 1 to determine whether the functions y_1 and y_2 are linearly dependent on the interval $(0, 1)$.

27. $y_1(t) = \cos t \sin t$, $y_2(t) = \sin 2t$
28. $y_1(t) = e^{3t}$, $y_2(t) = e^{-4t}$
29. $y_1(t) = te^{2t}$, $y_2(t) = e^{2t}$
30. $y_1(t) = t^2 \cos(\ln t)$, $y_2(t) = t^2 \sin(\ln t)$
31. $y_1(t) = \tan^2 t - \sec^2 t$, $y_2(t) \equiv 3$
32. $y_1(t) \equiv 0$, $y_2(t) = e^t$

33. Explain why two functions are linearly dependent on an interval I if and only if there exist constants c_1 and c_2 , not both zero, such that

$$c_1 y_1(t) + c_2 y_2(t) = 0 \quad \text{for all } t \text{ in } I.$$

34. **Wronskian.** For any two differentiable functions y_1 and y_2 , the function

$$(18) \quad W[y_1, y_2](t) = y_1(t)y_2'(t) - y_1'(t)y_2(t)$$

is called the *Wronskian*[†] of y_1 and y_2 . This function plays a crucial role in the proof of Theorem 2.

- (a) Show that $W[y_1, y_2]$ can be conveniently expressed as the 2×2 determinant

$$W[y_1, y_2](t) = \begin{vmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{vmatrix}.$$

- (b) Let $y_1(t), y_2(t)$ be a pair of solutions to the homogeneous equation $ay'' + by' + cy = 0$ (with $a \neq 0$) on an open interval I . Prove that $y_1(t)$ and $y_2(t)$ are linearly independent on I if and only if their Wronskian is never zero on I . [Hint: This is just a reformulation of Lemma 1.]

- (c) Show that if $y_1(t)$ and $y_2(t)$ are any two differentiable functions that are linearly dependent on I , then their Wronskian is identically zero on I .

35. **Linear Dependence of Three Functions.** Three functions $y_1(t)$, $y_2(t)$, and $y_3(t)$ are said to be linearly dependent on an interval I if, on I , at least one of these functions is a linear combination of the remaining two [e.g., if $y_1(t) = c_1 y_2(t) + c_2 y_3(t)$]. Equivalently (compare Problem 33), y_1, y_2 , and y_3 are linearly dependent on I if there exist constants C_1, C_2 , and C_3 , not all zero, such that

$$C_1 y_1(t) + C_2 y_2(t) + C_3 y_3(t) = 0 \quad \text{for all } t \text{ in } I.$$

Otherwise, we say that these functions are linearly independent on I .

[†]Historical Footnote: The Wronskian was named after the Polish mathematician H. Wronski (1778–1863).

For each of the following, determine whether the given three functions are linearly dependent or linearly independent on $(-\infty, \infty)$:

- (a) $y_1(t) = 1$, $y_2(t) = t$, $y_3(t) = t^2$.
 (b) $y_1(t) = -3$, $y_2(t) = 5 \sin^2 t$, $y_3(t) = \cos^2 t$.
 (c) $y_1(t) = e^t$, $y_2(t) = te^t$, $y_3(t) = t^2 e^t$.
 (d) $y_1(t) = e^t$, $y_2(t) = e^{-t}$, $y_3(t) = \cosh t$.

36. Using the definition in Problem 35, prove that if r_1, r_2 , and r_3 are distinct real numbers, then the functions $e^{r_1 t}$, $e^{r_2 t}$, and $e^{r_3 t}$ are linearly independent on $(-\infty, \infty)$. [Hint: Assume to the contrary that, say, $e^{r_1 t} = c_1 e^{r_2 t} + c_2 e^{r_3 t}$ for all t . Divide by $e^{r_2 t}$ to get $e^{(r_1 - r_2)t} = c_1 + c_2 e^{(r_3 - r_2)t}$ and then differentiate to deduce that $e^{(r_1 - r_2)t}$ and $e^{(r_3 - r_2)t}$ are linearly dependent, which is a contradiction. (Why?)]

In Problems 37–41, find three linearly independent solutions (see Problem 35) of the given third-order differential equation and write a general solution as an arbitrary linear combination of these.

37. $y''' + y'' - 6y' + 4y = 0$
 38. $y''' - 6y'' - y' + 6y = 0$
 39. $z''' + 2z'' - 4z' - 8z = 0$
 40. $y''' - 7y'' + 7y' + 15y = 0$
 41. $y''' + 3y'' - 4y' - 12y = 0$
42. (True or False): If f_1, f_2, f_3 are three functions defined on $(-\infty, \infty)$ that are pairwise linearly independent on $(-\infty, \infty)$, then f_1, f_2, f_3 form a linearly independent set on $(-\infty, \infty)$. Justify your answer.

43. Solve the initial value problem:

$$\begin{aligned} y''' - y' &= 0; & y(0) &= 2, \\ y'(0) &= 3, & y''(0) &= -1. \end{aligned}$$

44. Solve the initial value problem:

$$\begin{aligned} y''' - 2y'' - y' + 2y &= 0; \\ y(0) &= 2, & y'(0) &= 3, & y''(0) &= 5. \end{aligned}$$

45. By using Newton's method or some other numerical procedure to approximate the roots of the auxiliary equation, find general solutions to the following equations:

- (a) $3y''' + 18y'' + 13y' - 19y = 0$.
 (b) $y^{iv} - 5y'' + 5y = 0$.
 (c) $y^v - 3y^{iv} - 5y''' + 15y'' + 4y' - 12y = 0$.

46. One way to define hyperbolic functions is by means of differential equations. Consider the equation $y'' - y = 0$. The *hyperbolic cosine*, $\cosh t$, is defined as the solution of this equation subject to the initial values: $y(0) = 1$ and $y'(0) = 0$. The *hyperbolic sine*, $\sinh t$, is defined as the solution of this equation subject to the initial values: $y(0) = 0$ and $y'(0) = 1$.

- (a) Solve these initial value problems to derive explicit formulas for $\cosh t$ and $\sinh t$. Also show that $\frac{d}{dt} \cosh t = \sinh t$ and $\frac{d}{dt} \sinh t = \cosh t$.
 (b) Prove that a general solution of the equation $y'' - y = 0$ is given by $y = c_1 \cosh t + c_2 \sinh t$.
 (c) Suppose a, b , and c are given constants for which $ar^2 + br + c = 0$ has two distinct real roots. If the two roots are expressed in the form $\alpha - \beta$ and $\alpha + \beta$, show that a general solution of the equation $ay'' + by' + cy = 0$ is $y = c_1 e^{at} \cosh(\beta t) + c_2 e^{at} \sinh(\beta t)$.
 (d) Use the result of part (c) to solve the initial value problem: $y'' + y' - 6y = 0, y(0) = 2, y'(0) = -17/2$.

4.3 Auxiliary Equations with Complex Roots

The *simple harmonic equation* $y'' + y = 0$, so called because of its relation to the fundamental vibration of a musical tone, has as solutions $y_1(t) = \cos t$ and $y_2(t) = \sin t$. Notice, however, that the auxiliary equation associated with the harmonic equation is $r^2 + 1 = 0$, which has imaginary roots $r = \pm i$, where i denotes $\sqrt{-1}$.[†] In the previous section, we expressed the solutions to a linear second-order equation with constant coefficients in terms of exponential functions. It would appear, then, that one might be able to attribute a meaning to the forms e^{it} and e^{-it} and that these “functions” should be related to $\cos t$ and $\sin t$. This matchup is accomplished by Euler's formula, which is discussed in this section.

When $b^2 - 4ac < 0$, the roots of the auxiliary equation

$$(1) \quad ar^2 + br + c = 0$$

[†]Electrical engineers frequently use the symbol j to denote $\sqrt{-1}$.

Exercises 3.6, page 129

3. h	" e "
1	3
0.1	2.72055
0.01	2.71830
0.001	2.71828
0.0001	2.71828

7. x_n	y_n
1.1	0.10450
1.2	0.21668
1.3	0.33382
1.4	0.45300
1.5	0.57135

9. x_n	y_n
0.2	0.61784
0.4	1.23864
0.6	1.73653
0.8	1.98111
1.0	1.99705
1.2	1.88461
1.4	1.72447
1.6	1.56184
1.8	1.41732
2.0	1.29779

11. $\phi(1) \approx x(1; 2^{-3}) = 1.25494$

13. $\phi(1) \approx y(1; 2^{-3}) = 0.71698$

15. $x = 1.27$

17. x_n	$y_n(h = 0.2)$	$y_n(h = 0.1)$	$y_n(h = 0.025)$
0.1		-1	0.06250
0.2	-3	1	0.00391
0.3		-1	0.00024
0.4	9	1	0.00002
0.5		-1	0.00000
0.6	-27	1	0.00000
0.7		-1	0.00000
0.8	81	1	0.00000
0.9		-1	0.00000
1.0	-243	1	0.00000

We conclude that step size can dramatically affect convergence.

19. T_n			
Time	$K = 0.2$	$K = 0.4$	$K = 0.6$
Midnight	65.0000	65.0000	65.0000
4 A.M.	69.1639	68.5644	68.1300
8 A.M.	71.4836	72.6669	73.6678
Noon	72.9089	75.1605	76.9783
4 P.M.	72.0714	73.5977	74.7853
8 P.M.	69.8095	69.5425	69.2831
Midnight	68.3852	67.0500	65.9740

Exercises 3.7, page 139

$$1. y_{n+1} = y_n + h \cos(x_n + y_n) - \frac{h^2}{2} \sin(x_n + y_n) [1 + \cos(x_n + y_n)]$$

$$3. y_{n+1} = y_n + h(x_n - y_n) + \frac{h^2}{2}(1 - x_n + y_n) - \frac{h^3}{6}(1 - x_n + y_n) + \frac{h^4}{24}(1 - x_n + y_n)$$

5. Order 2, $\phi(1) \approx 1.3725$; order 4, $\phi(1) \approx 1.3679$

7. -11.7679 9. 1.36789 11. $x = 1.41$

13. $x = 0.50$

15. x_n	y_n
0.5	0.21462
1.0	0.13890
1.5	-0.02668
2.0	-0.81879
2.5	-1.69491
3.0	-2.99510

19. $v(3) \approx 0.24193$ with $h = 0.0625$

21. $z(1) \approx 2.87083$ with $h = 0.03125$

CHAPTER 4

Exercises 4.1, page 156

3. Both approach zero. 5. 0

7. $y(t) = -(30/61) \cos 3t - (25/61) \sin 3t$

9. $y(t) = -2 \cos 2t + (3/2) \sin 2t$

Exercises 4.2, page 164

1. $c_1 e^{t/2} + c_2 e^{-4t}$

3. $c_1 e^{-3t} + c_2 e^{-2t}$

5. $c_1 e^{-4t} + c_2 t e^{-4t}$

7. $c_1 e^{t/2} + c_2 e^{-2t/3}$

9. $c_1 e^{t/2} + c_2 t e^{t/2}$

11. $c_1 e^{-5t/2} + c_2 t e^{-5t/2}$

13. $3e^{-4t}$ 15. $\frac{4}{3}e^t - \frac{1}{3}e^{3t}$

17. $\left(\frac{7t}{3} + 2\right)e^{3t}$ 19. $e^{-t} - 2te^{-t}$

21. (a) $ar + b = 0$ (b) $ce^{-bt/a}$

23. $ce^{-4t/5}$ 25. $ce^{13t/6}$

27. Lin. dep. 29. Lin. indep. 31. Lin. dep.

33. If $c_1 \neq 0$, then $y_1 = -(c_2/c_1)y_2$.

35. (a) Lin. indep. (b) Lin. dep. (c) Lin. indep. (d) Lin. dep.

37. $c_1 e^t + c_2 e^{(-1-\sqrt{5})t} + c_3 e^{(-1+\sqrt{5})t}$

39. $c_1 e^{-2t} + c_2 t e^{-2t} + c_3 e^{2t}$

41. $c_1 e^{-3t} + c_2 e^{-2t} + c_3 e^{2t}$

43. $3 + e^t - 2e^{-t}$

45. (a) $c_1 e^{r_1 t} + c_2 e^{r_2 t} + c_3 e^{r_3 t}$ (where $r_1 = -4.832$, $r_2 = -1.869$, and $r_3 = 0.701$)

(b) $c_1 e^{r_1 t} + c_2 e^{-r_1 t} + c_3 e^{r_2 t} + c_4 e^{-r_2 t}$

(where $r_1 = 1.176$, $r_2 = 1.902$)

(c) $c_1 e^{-t} + c_2 e^t + c_3 e^{-2t} + c_4 e^{2t} + c_5 e^{3t}$

Exercises 4.3, page 172

1. $c_1 \cos 3t + c_2 \sin 3t$

3. $c_1 e^{3t} \cos t + c_2 e^{3t} \sin t$

5. $c_1 e^{-2t} \cos \sqrt{2}t + c_2 e^{-2t} \sin \sqrt{2}t$

7. $c_1 e^{-t/2} \cos(\sqrt{5}t/2) + c_2 e^{-t/2} \sin(\sqrt{5}t/2)$

9. $c_1 e^t + c_2 e^{t^2}$

11. $c_1 e^{-5t} + c_2 t e^{-5t}$
 13. $c_1 e^t \cos 5t + c_2 e^t \sin 5t$
 15. $c_1 e^{(3+\sqrt{53})t/2} + c_2 e^{(3-\sqrt{53})t/2}$
 17. $c_1 e^{t/2} \cos(3\sqrt{3}t/2) + c_2 e^{t/2} \sin(3\sqrt{3}t/2)$
 19. $c_1 e^t + c_2 e^{-t} \cos 2t + c_3 e^{-t} \sin 2t$
 21. $2e^{-t} \cos t + 3e^{-t} \sin t$
 23. $(\sqrt{2}/4)[e^{(2+\sqrt{2})t} - e^{(2-\sqrt{2})t}]$
 25. $e^t \sin t - e^t \cos t$
 27. $e^{2t} - \sqrt{2} e^t \sin \sqrt{2}t$
 29. (a) $c_1 e^{-t} + c_2 e^t \cos \sqrt{2}t + c_3 e^t \sin \sqrt{2}t$
 (b) $c_1 e^{2t} + c_2 e^{-2t} \cos 3t + c_3 e^{-2t} \sin 3t$
 (c) $c_1 \cos 2t + c_2 \sin 2t + c_3 \cos 3t + c_4 \sin 3t$
 31. (a) Oscillatory (b) Tends to zero
 (c) Tends to $-\infty$ (d) Tends to $-\infty$
 (e) Tends to $+\infty$
 33. (a) $y(t) = 0.3e^{-3t} \cos 4t + 0.2e^{-3t} \sin 4t$
 (b) $[\pi + \arctan(-1.5)]/4$
 (c) $2/\pi$
 (d) Decreases the frequency of oscillation, introduces the factor e^{-3t} , causing the solution to decay to zero
 35. $b \geq 2\sqrt{Ik}$
 37. (a) $c_1 \cos t + c_2 \sin t + c_3 t \cos t + c_4 t \sin t$
 (b) $(c_1 + c_2)e^{-t} \cos(\sqrt{3}t) + (c_3 + c_4t)e^{-t} \sin(\sqrt{3}t)$

Exercises 4.4, page 180

1. No 3. Yes 5. No 7. Yes
 9. $y_p \equiv -3$
 11. $y_p(x) = [(\ln 2)^2 + 1]^{-1} 2^x$
 13. $\cos 3t$
 15. $xe^x/2 + 3e^x/4$
 17. $-2t \cos 2t$
 19. $\left(\frac{t}{13} + \frac{8}{169}\right)te^{-3t}$ 21. $t^3 e^{2t}/6$
 23. $-\frac{1}{21}\theta^3 - \frac{1}{49}\theta^2 - \frac{2}{343}\theta$
 25. $e^{2t}(\cos 3t + 6 \sin 3t)$
 27. $(A_3 t^4 + A_2 t^3 + A_1 t^2 + A_0 t) \cos 3t$
 $+ (B_3 t^4 + B_2 t^3 + B_1 t^2 + B_0 t) \sin 3t$
 29. $e^t(A_4 t^4 + A_3 t^3 + A_2 t^2 + A_1 t + A_0)$
 31. $(A_3 t^4 + A_2 t^3 + A_1 t^2 + A_0 t)e^{-t} \cos t$
 $+ (B_3 t^4 + B_2 t^3 + B_1 t^2 + B_0 t)e^{-t} \sin t$
 33. $(1/5) \cos t + (2/5) \sin t$
 35. $\left(\frac{1}{10}t^2 - \frac{4}{25}t\right)e^t$

Exercises 4.5, page 185

1. (a) $5 \cos t$ (b) $\cos t - e^{2t}$ (c) $4 \cos t + 6e^{2t}$
 3. $y = -t + c_1 e^t + c_2 e^{-t}$
 5. $\theta = t - 1 + c_1 e^{2t} + c_2 e^{-t}$
 7. $y = \tan x + c_1 e^{\sqrt{2}x} + c_2 e^{-\sqrt{2}x}$
 9. Yes 11. No 13. Yes 15. No
 17. $y(t) = -t^2 + \frac{4t}{3} + \frac{1}{9} + c_1 e^{3t} + c_2 e^{-t}$

19. $y = (\cos x - \sin x)e^x/2 + c_1 e^x + c_2 e^{2x}$
 21. $y = (1/2)\theta e^{-\theta} \sin \theta + (c_1 \cos \theta + c_2 \sin \theta)e^{-\theta}$
 23. $y = e^t - 1$
 25. $z = e^{-x} - \cos x + \sin x$
 27. $y = -(3/10) \cos x - (1/10) \sin x - (1/20) \cos 2x$
 $+ (3/20) \sin 2x$
 29. $y = -(1/2) \sin \theta - (1/3)e^{2\theta} + (3/4)e^\theta + (7/12)e^{-\theta}$
 31. $y_p = (A_1 t + A_0)t \cos t + (B_1 t + B_0)t \sin t + C \cdot 10^t$
 33. $x_p(t) = (A \cos t + B \sin t)e^t + C_2 t^2 + C_1 t + C_0$
 $+ D_1 \cos 3t + D_2 \sin 3t + E_1 \cos t + E_2 \sin t$
 35. $y_p = (A_1 t + A_0) \cos 3t + (B_1 t + B_0) \sin 3t + C e^{5t}$
 37. $y_p = t^2 + 3t - 1$
 39. $y_p = (t/10 - 4/25)t e^t - 1/2$
 41. (a) $y_1 = -(2 \cos 2t + \sin 2t)e^{-t} + 2$
 for $0 \leq t \leq 3\pi/2$
 (b) $y_2 = y_h = (c_1 \cos 2t + c_2 \sin 2t)e^{-t}$
 for $t > 3\pi/2$
 (c) $c_1 = -2(e^{3\pi/2} + 1)$, $c_2 = -(e^{3\pi/2} + 1)$
 43. $y = -\cos t + (1/2) \sin t - (1/2)e^{-3t} + 2e^{-t}$
 45. (a) $y(t) = \frac{2V \cos(\pi/2V)}{V^2 - 1} \sin t$ for $V \neq 1$;
 $y(t) = \frac{\pi}{2} \sin t$ for $V = 1$
 (b) $V \approx 0.73$
 47. (a) $2 \sin 3t - \cos 6t$ (b) No solution
 (c) $c \sin 3t - \cos 6t$, where c is any constant

Exercises 4.6, page 191

1. $c_1 \cos 2t + c_2 \sin 2t$
 $- (1/4)(\cos 2t) \ln |\sec 2t + \tan 2t|$
 3. $c_1 e^t + c_2 t e^t + t e^t \ln |t|$
 5. $c_1 \cos 4\theta + c_2 \sin 4\theta + (\theta/4) \sin 4\theta$
 $+ (1/16)(\cos 4\theta) \ln |\cos 4\theta|$
 7. $(2 \ln t - 3)t^2 e^{-2t}/4 + c_1 e^{-2t} + c_2 t e^{-2t}$
 9. $y_p = -2t - 4$
 11. $y = -(\cos t) \ln |\sec t + \tan t| + (1/10)e^{3t}$
 $- 1 + c_1 \cos t + c_2 \sin t$
 13. $c_1 \cos 2t + c_2 \sin 2t + (1/24) \sec^2 2t - 1/8$
 $+ (1/8)(\sin 2t) \ln |\sec 2t + \tan 2t|$
 15. $c_1 \cos t + c_2 \sin t - t^2 + 3 + 3t \sin t$
 $+ 3(\cos t) \ln |\cos t|$
 17. $c_1 \cos 2t + c_2 \sin 2t - e^t/5$
 $- (1/2)(\cos 2t) \ln |\sec 2t + \tan 2t|$
 19. $y = e^{1-t} - e^{t-1} + \frac{e^t}{2} \int_1^t \frac{e^{-u}}{u} du - \frac{e^{-t}}{2} \int_1^t \frac{e^u}{u} du$
 $[y(2) \approx -1.93]$
 21. 0.3785
 23. $c_1 e^t + c_2(t+1) - t^2$
 25. $c_1(5t-1) + c_2 e^{-5t} - t^2 e^{-5t}/10$

Exercises 4.7, page 199

1. Unique solution on $(-\pi/2, \pi/2)$
 3. Unique solution on $(0, \infty)$
 5. Does not apply; $t = 0$ is a point of discontinuity